CARLOS ALBERTO DE LA PEÑA GONZÁLEZ

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EDUCATION

University of Michigan – Dearborn Dearborn, MI
BSE Computer Engineering & BSE Electrical Engineering (Dual Degree) Expected May 2029
Cumulative GPA: 3.38/4.0 | Relevant Coursework: Digital Systems, Calculus I & II, General Physics, Intro to Engineering & Design

WORK EXPERIENCE

Tiger Price | Remote
Lead Full-Stack Developer – E-Commerce Platform, Jul 2025 – Aug 2026
- Sole developer of a full e-commerce platform for a grocery delivery marketplace: storefront, catalog, cart, checkout, and admin dashboard, built with React, Vite, and CSS
- Integrated Firebase for real-time database, authentication, and inventory, with security rules scoping every read and write to its owner
- Implemented Stripe payments for secure end-to-end transactions and server-enforced cancellation windows

KinPOS | Colombia
Technology Intern – Point of Sale Systems, Apr 2024
- Gained hands-on experience with POS systems and retail technology, assisting with setup, configuration, and support of POS hardware and software

PCPros Onsite | Michigan
Computer Repair Technician Intern, Jun 2023 – Jul 2023
- Diagnosed and repaired hardware and software issues on laptops and desktop PCs for residential customers
- Performed Windows installs, maintenance, data recovery, and hardware upgrades, translating complex issues into clear solutions for customers

PROJECTS & ENTREPRENEURSHIP

Elementa – Elemental Dice Roguelite | Solo Game Designer & Developer
Browser Game | React, Framer Motion, Canvas, Web Audio | 2026 (playable alpha, v0.9)
- Designed and built a full roguelite from a living design doc: 140+ dice, 62 relics, 15 loadouts, a 45-round campaign across four realms, three story paths, and nine endings
- Engineered a pure, data-driven scoring engine that itemizes every point, plus adjacency reactions and drag-to-reorder dice; shipped as numbered alpha releases with bilingual EN/ES UI
- Generated all art and music procedurally in code (Canvas, Web Audio); verified with Node tests and an AI-assisted workflow (Claude Code)

Ohmstead – Electronics Factory Builder | Solo Game Designer & Developer
Isometric Factory Game | PixiJS, JavaScript | 2026 (in development)
- Building a Factorio-style game where players manufacture electronics from raw ore to a working desktop computer, drawing on my electrical and computer engineering studies
- Built a deterministic 20 Hz simulation engine, kept separate from the renderer, with two-lane belts, drills, recipe machines, crafting, research labs, and save/load verified by state hash
- Authored a 57-node tech tree, 39 missions, and 72 recipes as data; drawing isometric machine art in code with PixiJS; 11 Node test suites

Project Wellness – Fitness App | Solo Developer
Mobile App | Flutter, Dart, SQLite, OCR | In active development
- Built a local-first fitness app with no backend or accounts by design, so users keep full control of their data, with manual export and full backup
- Implemented barcode and OCR nutrition-label scanning, a workout tracker with exercise library, and "Body Rank," a gamified system leveling 13 muscle groups through 9 tiers

Tootor – On-Demand Tutoring App | Co-Founder & Lead Developer
Mobile App Startup | iOS & Android (Flutter) | Jan 2025 – Jun 2025
- Co-founded a startup building an Uber-style marketplace connecting students with online tutors; led all product design and development decisions
- Developed the full cross-platform app in Flutter, including in-app purchases and the tutor booking/payment flow; reached near-launch before halting due to funding constraints

Line-Following Robot | University of Michigan – Dearborn
ENGR 100 – Intro to Engineering & Design | Team Project | Fall 2025, ranked first in lab section
- Designed the 3D chassis in CAD, programmed the Arduino for line detection and autonomous navigation, and integrated the hardware with custom firmware as a team

TECHNICAL SKILLS

Languages & Frameworks: JavaScript, React, Tailwind CSS, Framer Motion, HTML/CSS, Dart (Flutter), C++ (Arduino), SQLite
Tools & Platforms: Vite, Firebase, Stripe, Git/GitHub, PixiJS, Three.js, Canvas & Web Audio APIs, Node.js testing, Arduino IDE, CAD (3D Modeling)
AI & Design Tools: Claude Code, ChatGPT, Gemini, Figma (learning)
Hardware & Systems: PC assembly & repair, Windows OS, embedded systems (Arduino)
Spoken Languages: Spanish (Native), English (Fluent, ESL Certified), French (A2/B1)

ADDITIONAL

Certifications: ESL | **Studied & worked in:** Mexico, USA, Colombia, France